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Relationship Between Arctic Oscillation and PM2.5 Concentrations in Seoul: An Analysis Using ERA5 Data and WRF Model Assessments

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1. Introduction

1. Monitoring and predicting **fine particulate matter (PM2.5)** concentrations is crucial for public health, especially in urban areas with significant air pollution risks.
2. Effective air quality management in densely populated and industrial regions like East Asia requires understanding factors influencing PM2.5 variability, including **synoptic-field changes, interannual and seasonal variability, and climatic indices**.
3. The **Arctic Oscillation (AO)** significantly affects regional weather patterns, potentially impacting urban air quality in cities like Seoul.
4. This study investigates the relationship between PM2.5 concentrations in Seoul and the AO using ECMWF Reanalysis v5 (ERA5) data and the **Weather Research and Forecasting (WRF) model**.
5. Daily average PM2.5 concentrations from **48 air quality stations in Seoul from 2015 to 2023** were analyzed, focusing on the **November to April period**, which typically experiences numerous high concentration events.
6. **Regression analysis** reveals the impact of AO phases on Seoul's air quality, highlighting the **correlation** between PM2.5 concentrations and various weather variables during positive and negative AO phases.
7. The study evaluated the **WRF model's performance** in predicting weather variables during high pollution events to enhance predictive accuracy and preparedness.
8. The research confirms the AO's impact on Seoul's air quality, emphasizing the need to consider synoptic-scale weather variables and climate indices, particularly the AO phase, in forecasting local-scale PM2.5 concentrations in East Asia.

2. Data & Methods

• Selection of High PM2.5 Concentration Date

- Korean PM2.5 Hourly Standards

Good	0-15 μg/m³
Moderate	16-35 μg/m³
Unhealthy	36-75 μg/m³
Very Unhealthy	76 μg/m³ ≤

- Statistics

Good	18.3% (276 days)
Moderate	58.0% (875 days)
Unhealthy	22.3% (336 days)
Very Unhealthy	1.5% (22 days)

- Analysis Standards
  - Analysis Period: 2015-01-01 ~ 2023-02-28 (Only NDJFMA, 50 months, 1509 days)
  - Analysis station: Seoul, Korea, South. 40 stations
  - High concentration date: Daily average PM2.5 concentration > 35 μg/m³ (Unhealthy, Very Unhealthy)
    * National ambient air quality standard: 24-hour 35 μg/m³ or less
    * Hourly PM2.5 Concentration data is given by AirKorea, operated by Ministry of Environment
  - Total **358 date (23.7%) of high PM2.5 concentration** in Seoul, Korea, South

• AO Phase Index

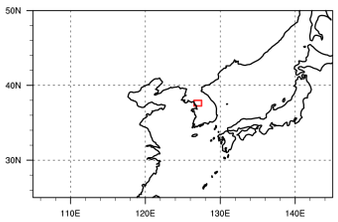
- Daily AO Index value is given by Climate Prediction Center of National Centers for Environmental Prediction
- AO Index up to the day before is provided and can be used
- **1 day lag AO Index value** and PM2.5 concentration relationship has been analyzed
- AO Index standard deviation: 1.012 (Monthly Averaged from 1950 to 2023)
- 0.5 * std = 0.506 is used for phase classification

Positive (> 0.506)	46.6% (167 days)
Neutral	27.9% (100 days)
Negative (< - 0.506)	25.4% (91 days)

• Weather Data

	ECMWF Reanalysis v5 (ERA5)	The Integrated Multi-satellite Retrievals for GPM (IMERG)
Temporal Resolution	6 hourly	30 minutes
Spatial Resolution	0.25° x 0.25°	0.1° x 0.1°
Variables	10m u / v wind, 2m temperature / Specific humidity, Planetary Boundary Layer (PBL) height	Precipitation

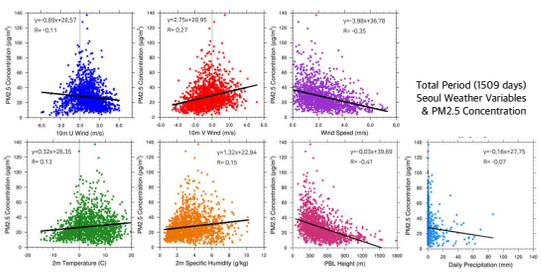
• Analysis Domain



< East Asia and Seoul, Korea, South (Red rectangle) >

3. Results

• Regression with ERA5 Weather Variables & PM2.5 Concentration



Total Period (1509 days) Seoul Weather Variables & PM2.5 Concentration

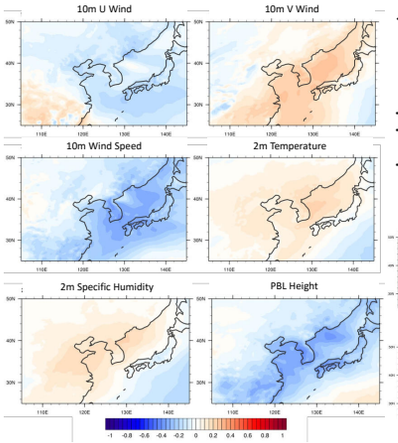
◆ **Proportional relationship observed with:**
- Surface north-south wind
- Surface temperature
- Surface humidity

◆ **Inverse relationship observed with:**
- Surface east-west wind
- Wind speed
- PBL height
- Daily accumulated precipitation

◆ **PM2.5 concentration increases when:**
- Surface wind is more southerly or easterly
- Wind speed is weaker
- Surface temperature is higher
- Surface humidity is higher

◆ **PM2.5 concentration decreases when:**
- PBL height is higher

• Correlation between Weather Variable & Seoul PM2.5 Concentration



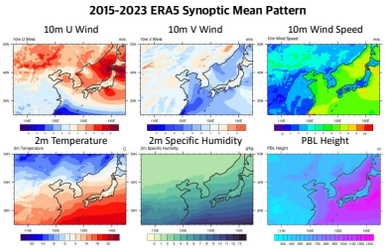
❖ **Seoul's Air Quality:**

◆ **Negative correlation with:**
- 10m U wind
- 10m wind speed
- PBL height

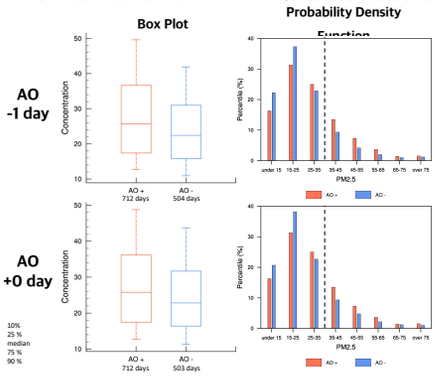
◆ **Positive correlation with:**
- 10m V wind
- 2m temperature
- 2m humidity

- Wind variables show a high correlation with air quality over the sea.
- PBL height shows a high correlation with air quality over both land and sea, centered around the Korean Peninsula.
- Temperature and humidity show a positive correlation with air quality not only in the Korean Peninsula but also in China.

2015-2023 ERA5 Synoptic Mean Pattern



• AO Index Correlation with PM2.5 Concentrations in Seoul



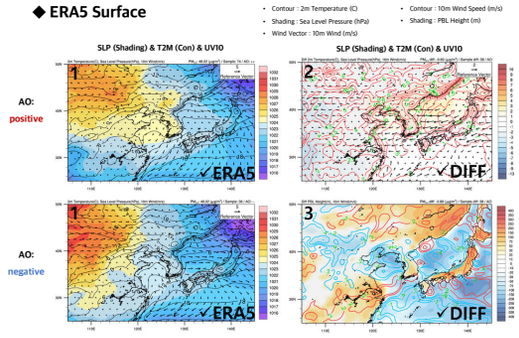
- PM2.5 concentration values were higher during positive AO phases compared to negative AO phases
- The difference between AO phases was more pronounced compared to the ENSO index

◆ **Probability density function (PDF) analysis**
- Higher frequency of "good" air quality days during negative AO phases
- Higher frequency of days with PM2.5 concentrations above 25 μg/m3 during positive AO phases

- PM2.5 concentrations were generally higher during positive AO phases compared to negative AO phases
- This trend was more distinct compared to the ENSO index
- The observed trend was consistent whether analyzing the AO index with a one-day lead or on the same day.

• Impact of AO Phases on PM2.5 Levels and WRF Model Predictions in Seoul

◆ **ERA5 Surface**



❖ **Analysis Target**

◆ **Analysis Period:** November to April (2019-2023, up to February 2023)

◆ **AO Phases Breakdown:**
- Positive Phase: 320 days (74 high PM2.5 days)
- Neutral Phase: 103 days (18 high PM2.5 days)
- Negative Phase: 241 days (36 high PM2.5 days)

❖ **Key Findings on AO Phase Characteristics**

◆ **Surface Pressure:**
- No significant differences across AO phases.

◆ **Surface Temperature (2m):**
- Higher temperatures during positive AO phases.

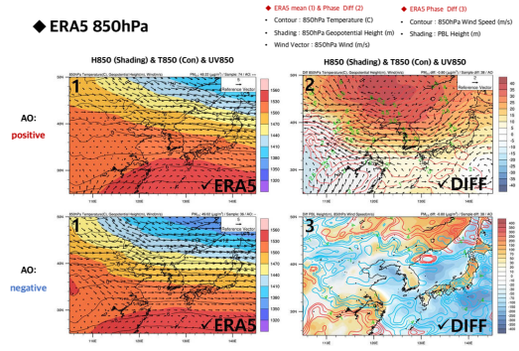
◆ **Surface Wind (10m):**
- Stronger westerly winds over Yellow Sea and East Sea during negative AO phases.
- Lower wind speeds over Yellow Sea and inland China during positive AO phases, with corresponding lower PBL Heights.

◆ **PBL Height in Seoul:**
- Higher during positive AO phases despite weaker winds, likely influenced by higher surface temperatures

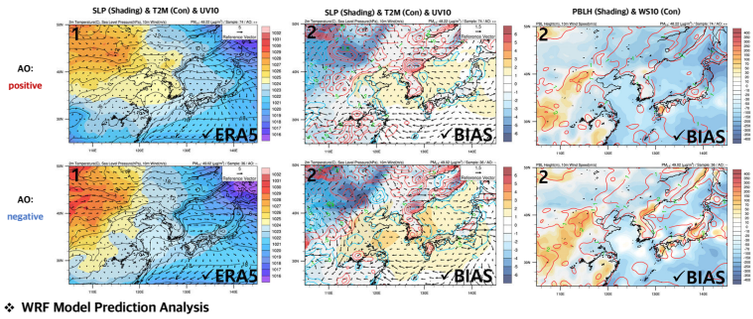
◆ **850hPa Level:**
- Stronger winds during negative AO phases.
- Higher PBL Height during positive AO phases due to higher temperatures.

◆ **Implication:**
- Temperature-induced thermal mixing affecting PBL Height should be considered in air quality prediction.

◆ **ERA5 850hPa**



◆ **WRF Bias (F108)**



❖ **WRF Model Prediction Analysis**

◆ **Systematic Errors:**
- Warm bias over land regardless of AO phase.
- High sea level pressure around Korea and low sea level pressure over North China Plain.

◆ **Surface Temperature and Wind:**
- Higher than ERA5 in Korea regardless of AO phase.

◆ **PBL Height:**
- Positive(Negative) AO Phase: Negative(Positive) bias.
- More days of high PM2.5 with lower PBL Height during negative AO phases.
- Systematic errors with higher surface winds and temperatures, leading to lower PBL Heights, but during negative AO phases, WRF models PBL Height even higher.

◆ **Prediction Difficulty:**
- Greater challenges in predicting air quality during negative AO phases due to these discrepancies.

4. Summary

- **Analysis of PM2.5 concentrations in Seoul** from 2015 to 2023, focusing on high concentration events during the November to April period, revealed a significant influence of **the Arctic Oscillation (AO)** on urban air quality.
- **Regression analysis** demonstrated distinct correlations between PM2.5 concentrations and various weather variables during positive and negative **AO phases**, emphasizing the importance of synoptic-scale weather patterns in determining local-scale pollution levels.
- Evaluation of the **Weather Research and Forecasting (WRF) model's performance** during high pollution events highlighted areas for improvement in predictive accuracy, particularly regarding synoptic-scale weather variables and the AO phase.
- The study confirms **the AO's impact on Seoul's air quality** and underscores the necessity of incorporating **synoptic-scale weather variables** and **climate indices**, such as the **AO phase**, into air quality forecasting models for East Asia.
- Additionally, **CMAQ(Community Multiscale Air Quality model)** results will be analyzed to examine the relationship between observed and model PM2.5 concentration data in relation to the **Arctic Oscillation (AO)** climate index, and **WRF model output variables.**

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